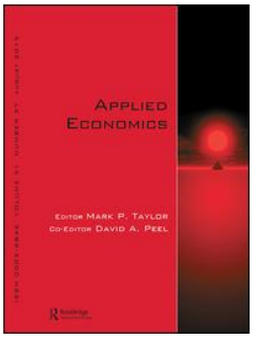

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Demand elasticities for 9 goods in 37 countries

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ABSTRACT

In most OECD countries, consumption absorbs more than three fourths of their country's GDP. Therefore, the study of their consumption patterns is important. This paper gives up-to-date estimates of income and price elasticities for 9 commodities in each of 37 countries that are members (or affiliates) of the OECD. In addition, the paper presents a detailed analysis of OECD consumption patterns in the form of investigating empirical regularities and testing demand theory hypotheses. Using differential demand equations, we find that the hypothesis of homogeneity hypothesis is supported in all countries; there is some evidence that symmetry is problematic, although we cannot be sure about this; and preference independence (whereby the marginal utility of each good depends only on its consumption) is not rejected in 90% of the countries. The study also finds that income elasticities of rich countries are generally lower than those of poorer countries, and, in general, there is a negative linear relationship between own-price elasticities and income. These elasticities can be used as key inputs in economic policy modelling such as CGE modelling and GST estimation.

KEYWORDS
Consumption; demand elasticities; differential demand equations; preference independence; OECD

JEL CLASSIFICATION
C30; D12

I. Introduction

The Organization for Economic Co-operation and Development (OECD) plays an important role in world affairs, since its inception 50 years ago, in a number of areas including climate change, trade and investment openness and sustainable development. The important contribution OECD countries make to the world economy is evident from an increasing number of non-members seeking membership. Recently, Chile, Estonia, Israel, Latvia and Slovenia have been admitted as OECD member countries. Further countries such as Brazil, China, India, Indonesia and South Africa were recently allowed to participate in some of the main activities of the OECD with the expectation of obtaining OECD membership in the near future.

Most OECD member countries are considered to be highly industrialized developed countries. Total consumption in most of these countries absorb more than three fourths of the country's GDP and is the largest of the macroeconomic aggregates, thus having great significance for the state of the economy as a whole and business conditions. The quality of life of the people in these countries are heavily dependent on their consumption patterns of goods such as food, clothing, housing, education, transport etc. For policy analysis purposes, knowing the consumer reaction on the purchase of consumer goods due to income and price changes are crucial. The purpose of this study is to analyse the current consumption patterns of the OECD consumers using more recent consumption expenditure data. The income and price elasticities resulting from this study for nine consumer goods purchased by consumers in the 37 OECD countries can be used as key inputs in economic policy modelling such as public-utility pricing, CGE modelling and GST estimation. In addition we will also compare the results of this study with previous studies to see any patterns of change in the consumption of consumer goods in recent years. While there are a number of previously published cross-country studies available, to the best of our knowledge, the latest detailed study published

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on OECD consumption patterns is Selvanathan and Selvanathan (2003). Selvanathan and Selvanathan used data for only 23 OECD countries covering only the period up to 1996.[1] Since then, the number of member countries have increased to 35 and more recent data has now become available, new estimates of income and price elasticities are long overdue. The current study aims to fill that gap by modelling the consumption patterns of 9 commodities in each of the 37 countries that are members (or affiliates) of the OECD.

The organization of the paper is as follows. In Section 2, we present a summary of the OECD consumption expenditure data and in Section 3, we present a preliminary data analysis of the data. In Section 4 we present the model for estimation, the results on testing the demand theory hypotheses, the estimation results. Section 5 reports the implied income and price elasticities and a comparison of our results with an older one. We present an analysis of the income and own-price elasticity estimates in Section 6. Finally, in Section 7 we present our concluding comments.

II. The OECD data

In this study, we have used the basic data consisting of annual consumption expenditure (in current and constant 2010 prices) for 37 OECD countries[2] for which data are available on the OECD website (online)[3]. While the data period ranges from 1959 to 2015 for some countries, for a majority of them, the sample period is from 1995 to 2015. While on the OECD data base generally commodity groups were classified into 12 categories, due to data unavailability for some of these groups for a few countries, we combine these 12 groups into 9 groups. We have combined 'Food and non-alcoholic beverages' with 'Alcoholic beverages, tobacco and narcotics' and, 'Transport' and 'Communication' have been moved to 'Miscellaneous goods and services'. Table 1 gives the description of each of the 9 commodity groups considered in this study.[4]

Table 1. Classification of commodities.

	Commodity group	Description
1	Food	Food, non-alcoholic and alcoholic beverages and narcotics
2	Clothing	Clothing and footwear
3	Housing	Housing and utilities
4	Durables	Furniture, furnishings, household equipment and operation
5	Health	Medical care and health expenses
6	Recreation	Recreation and cultural activities
7	Education	Educational activities
8	Restaurants	Restaurants and hotels
9	Miscellaneous	All other goods and services not included in 1-8

Table 2 lists the names of the 37 countries covered in this study (Column 1), data period (Column 2), number of observations (Column 3), per capita GDP for the year 2010 in international dollars and indexed as US = 100 (Columns 4 and 5) and food and alcohol share in 2010 (%) in column 6. The order of the list of countries is in descending order of the GDP per capita. As can be seen, per capita GDP in 2010 across the OECD countries differ widely (with US = 100), from 22.1 for Columbia to 177.3 for Luxembourg. The food budget share (in 2010) also varies widely between 9.0% for the US and 31.7% for Lithuania. As can be noted from columns (5) and (6), in general, as per capita GDP decreases, food budget share tends to increase.

III. Preliminary data analysis of OECD consumption patterns

We denote p_{it} and q_{it} for the price and the per capita quantity consumed of good i (i = 1,2, …, n) during year t (t = 1,2, …, T). Therefore, the expenditure on commodity i in period t will be $p_{it}q_{it}$ and the total expenditure M_t on all n commodities in period t is

$$M_t = \sum_{i=1}^{n} p_{it}q_{it}$$

[1]For other previous studies, see Chen (1999), Clements and Theil (1979), Cranfield et al. (2000), Gao (2011), Goldberger and Gamaletsos (1970), Lluch and Powell (1975), Lluch, Powell, and Williams (1977), Meade et al. (2014), Muhammad et al. (2011), Parks and Barten (1973), Pollak and Wales (1987), Regmi and Seale (2010), Rimmer and Powell (1992), Seale and Regmi (2006), Seale, Regmi, and Bernstein (2003), Seale et al. (2014), Theil and Clements (1987), Theil, Chung, and Seale (1989) and Theil, Suhm, and Meisner (1981).

[2]There are currently 36 OECD member nations. Due to data unavailability Chile is not included in this study and two other current non-OECD countries (South Africa and Colombia) whose data are available in the OECD database are included. These two countries, even though non-members work very closely with the OECD. So, in total, we consider 37 OECD countries and their affiliates.

[3]Source: https://www.oecd-ilibrary.org/economics/data/oecd-national-accounts-statistics_na-data-en.

[4]Further details about the extraction of the OECD data are presented in an Appendix to this paper, available from the corresponding author.

Table 2. The OECD data.

Country (1)	Period (2)	Observations (3)	Per capita GDP, 2010 Int. $'s (4)	Per capita GDP, 2010 US = 100 (5)	Food and alcohol share in 2010 (%) (6)
1. Luxembourg	1995 – 2015	21	85,621	177.3	18.0
2. Norway	1970 – 2014	45	58,054	120.2	16.5
3. Switzerland	1995 – 2013	19	53,132	110.0	12.9
4. United States	1970 – 2015	46	48,291	100.0	9.0
5. Netherlands	1995 – 2014	20	44,542	92.2	14.3
6. Denmark	1995 – 2015	21	43,057	89.2	15.1
7. Ireland	1996 – 2014	19	42,803	88.6	16.1
8. Australia	1985 – 2014	30	42,350	87.7	13.8
9. Austria	1995 – 2015	21	41,944	86.9	13.3
10. Sweden	1993 – 2015	23	41,649	86.2	15.8
11. Belgium	1995 – 2014	20	40,003	82.8	16.8
12. Germany	1995 – 2014	20	39,918	82.7	13.5
13. Canada	1981 – 2015	35	39,885	82.6	13.0
14. Finland	1975 – 2015	41	38,781	80.3	17.1
15. Iceland	1995 – 2014	20	38,411	79.5	18.9
16. France	1959 – 2015	57	37,209	77.1	16.3
17. UK	1995 – 2015	21	35,769	74.1	12.7
18. Japan	1994 – 2013	20	35,203	72.9	16.5
19. Italy	1995 – 2014	20	34,893	72.3	18.4
20. Spain	1995 – 2014	20	31,968	66.2	16.6
21. New Zealand	1987 – 2014	28	31,133	64.5	19.9
22. Korea	1970 – 2015	46	30,664	63.5	15.0
23. Israel	1995 – 2014	20	29,646	61.4	18.2
24. Greece	1995 – 2015	21	28,061	58.1	20.3
25. Slovenia	1995 – 2015	21	27,740	57.4	19.8
26. Czech Rep.	1995 – 2015	21	27,636	57.2	22.1
27. Portugal	1995 – 2015	21	27,331	56.6	19.0
28. Slovak Rep.	1995 – 2015	21	24,939	51.6	22.2
29. Estonia	1995 – 2015	21	21,613	44.8	28.8
30. Hungary	1995 – 2015	21	21,435	44.4	24.2
31. Poland	1995 – 2015	21	20,798	43.1	27.0
32. Lithuania	1995 – 2014	20	19,965	41.3	31.7
33. Latvia	1995 – 2014	20	17,652	36.6	30.5
34. Turkey	1998 – 2014	17	17,464	36.2	27.0
35. Mexico	2003 – 2014	12	14,586	30.2	25.4
36. South Africa	1975 – 2014	40	11,652	24.1	24.8
37. Colombia	2000 – 2014	15	10,680	22.1	22.0

Notes: Data on GDP and food and alcohol share are from OECD website. Chile is also an OECD member nation. However, reliable data are unavailable. Columbia and South Africa, though non-OECD members, work very closely with the OECD and are considered part of the OECD and, therefore, are included in the current study.

and the proportion of total expenditure allocated to commodity *i*, (also known as the *budget share of commodity i)*, is

$$w_{it} = \frac{p_{it}q_{it}}{M_t}$$

Table 3 presents the mean budget shares of the 9 commodity groups averaged over the corresponding sample period for each country. The last 5 rows of the table present the mean, minimum, maximum, standard deviation (SD) and the coefficient of variation (CV) of the average budget shares for each commodity across countries. As can be seen, the average budget share for food in the OECD countries is 20.5 percent (in the range 11.6–34.7), clothing 5.3 percent (range 3.0–7.7), housing 20.9 percent (range 13.-0–27.9), durables 5.7 percent (range 3.1–9.2), health 4.1 percent (range 1.5–15.6), recreation 8.5 percent (range 4.2–11.4), education 1.4 percent (range 0.2–5.4), restaurants 7.4 percent (range 2.8–16.4) and miscellaneous 26.1 percent (range 19.7–30.9). Across all OECD countries as a single commodity, the largest average budget share is held by food and housing which is each about at one-fifth of total income. As can be seen from the coefficient of variation (CV), the three commodities with very high variation in their budget shares are education (79.4%), health (64.3%) and restaurants (40.7%) and the ones with the least variation are housing (16.2%), durables (20.3%) and clothing (21.4%)[5].

[5]In an Appendix to this paper, available from the corresponding author, we present the budget shares for each country during various sub-sample periods.

Table 3. Budget shares (Means × 100).

	Country (1)	Food (2)	Clothing (3)	Housing (4)	Durables (5)	Health (6)	Recreation (7)	Education (8)	Restaurants (9)	Miscellaneous (10)
1	Australia	14.5	4.2	20.7	5.6	5.2	10.8	3.2	7.0	28.7
2	Austria	13.7	6.4	20.7	6.9	3.7	10.5	0.7	11.7	25.8
3	Belgium	17.1	5.0	23.7	6.1	5.6	9.2	0.4	5.6	27.2
4	Canada	14.5	5.3	23.7	6.1	3.4	9.0	1.1	7.3	29.6
5	Colombia	22.2	6.7	16.6	4.5	3.2	4.9	4.9	11.2	25.7
6	Czech Republic	24.1	4.1	24.5	5.6	2.0	10.2	0.6	8.0	20.9
7	Denmark	15.9	4.5	27.3	5.3	2.7	11.3	0.8	5.5	26.8
8	Estonia	28.4	6.6	19.7	4.6	2.8	8.1	0.9	6.6	22.2
9	Finland	21.8	5.1	22.5	5.3	3.4	10.9	0.4	6.6	24.0
10	France	20.5	7.3	20.5	6.6	3.0	8.7	0.7	6.0	26.7
11	Germany	14.3	5.4	24.2	7.0	4.4	9.6	0.7	5.0	29.4
12	Greece	20.7	5.0	19.0	4.4	4.5	4.6	2.1	13.7	26.0
13	Hungary	25.5	3.5	19.3	5.5	4.0	7.9	1.5	6.5	26.3
14	Iceland	19.2	5.0	19.9	6.5	3.0	11.3	1.2	8.2	25.6
15	Ireland	16.9	5.1	20.2	6.0	3.5	7.0	2.3	13.9	25.2
16	Israel	18.1	3.4	23.0	6.9	3.2	7.0	2.7	6.1	29.7
17	Italy	19.2	7.0	20.7	7.1	3.3	7.1	1.0	8.9	25.7
18	Japan	17.4	4.3	23.7	4.3	3.8	10.4	2.2	6.8	27.0
19	Korea	26.5	7.7	14.6	4.0	3.7	6.4	5.4	8.4	23.4
20	Latvia	32.2	5.1	20.2	3.1	4.0	8.4	1.8	5.6	19.7
21	Lithuania	34.7	6.4	16.2	5.4	4.2	6.4	0.6	2.9	23.1
22	Luxembourg	18.7	5.2	21.6	6.6	1.9	7.1	0.7	7.3	30.9
23	Mexico	25.0	3.0	21.4	5.7	4.4	4.2	1.5	4.4	30.3
24	Netherlands	14.2	5.4	20.3	6.3	3.9	11.4	0.7	7.2	30.6
25	New Zealand	19.5	4.5	25.5	6.1	2.2	10.1	0.9	5.8	25.5
26	Norway	21.3	6.7	21.3	6.7	2.4	10.6	0.5	5.2	25.6
27	Poland	28.7	4.9	20.9	4.5	4.0	7.9	1.2	2.9	25.0
28	Portugal	20.2	6.4	15.0	6.1	4.7	7.3	1.2	10.7	28.4
29	Slovak Republic	25.7	4.8	23.8	5.5	3.0	8.8	1.2	6.6	20.6
30	Slovenia	21.0	5.6	18.9	5.7	3.4	9.8	1.1	6.7	27.8
31	South Africa	28.6	6.2	13.0	9.2	5.8	5.5	2.1	2.8	26.7
32	Spain	17.9	5.4	18.5	5.2	3.4	8.1	1.6	16.4	23.5
33	Sweden	16.5	4.7	27.9	4.7	3.0	11.3	0.2	5.2	26.5
34	Switzerland	13.4	3.6	24.1	4.2	12.8	9.7	0.5	7.5	24.2
35	Turkey	26.7	6.9	17.8	7.6	3.5	4.4	1.1	6.2	25.7
36	UK	12.9	5.3	25.4	5.1	1.5	10.5	1.3	9.4	28.6
37	United States	11.6	5.2	18.8	5.5	15.6	8.6	1.8	6.6	26.4
	Mean	20.5	5.3	20.9	5.7	4.1	8.5	1.4	7.4	26.1
	Minimum	11.6	3.0	13.0	3.1	1.5	4.2	0.2	2.8	19.7
	Maximum	34.7	7.7	27.9	9.2	15.6	11.4	5.4	16.4	30.9
	Std dev	5.7	1.1	3.4	1.2	2.6	2.1	1.1	3.0	2.7
	CV	27.8	21.4	16.2	20.3	64.3	24.9	79.4	40.7	10.5

To summarize the consumption data, we use the Divisia (1925) indices. Divisia price and volume (quantity) indices that measure the overall movements in prices and consumption are defined as:

$$DP_t = \sum_{i=1}^{9} \bar{w}_{it} Dp_{it}, \qquad DQ_t = \sum_{i=1}^{9} \bar{w}_{it} Dq_{it},$$

where $\bar{w}_{it} = {}^1\!/_2\left(w_{it} + w_{i,t-1}\right)$ is the arithmetic average of the ith budget share; $Dp_{\mathrm{it}} = ln\ p_{\mathrm{it}} - ln\ p_{\mathrm{it\text{-}1}}$ and $Dq_{\mathrm{it}} = ln\ q_{\mathrm{it}} - ln\ q_{\mathrm{it\text{-}1}}$, respectively, are the price and quantity log-changes (when multiplied by 100, these log-changes can be interpreted as percentage changes or percentage growth rates from year t–1 to year t). Here and elsewhere, *ln* refers to the natural logarithm. It can be shown that the volume index DQ_t can also be interpreted as the change in real income of the consumer. The indices above are weighted first-order moments of the price and quantity changes and measure the centre of gravity of the price and consumption distributions. The corresponding second-order moments are the Divisia price and quantity variances:

$$V_t^p = \sum_{i=1}^{9} \bar{w}_{it} (Dp_{it} - DP_t)^2,$$

$$V_t^q = \sum_{i=1}^{9} \bar{w}_{it} (Dq_{it} - DQ_t)^2.$$

These measure the dispersion of the price and quantity distributions. There is also the price-quantity covariance and correlation defined as

$$V_t^{pq} = \sum_{i=1}^{9} \bar{w}_{it}(Dp_{it} - DP_t)(Dq_{it} - DQ_t),$$

$$\rho_t = \frac{V_t^{pq}}{\sqrt{V_t^p V_t^q}}.$$

As the demand curves slope down, we expect both V^{pq} and ρ to be negative.

Table 4 presents the Divisia price and quantity indices, variances and correlations for each country, averaged over their respective sample periods. As can be seen, while prices in the OECD countries have grown at a rate of 3.45 percent per annum, consumption have increased at a rate of 2.13 percent per annum. Figure 1 presents the plot of quantity standard deviation (square root of column 5) against price standard deviation (square root of column 4) with a 45-degree line. As can be seen, quantity standard deviation generally exceeds price standard deviation. That is, quantity changes are more dispersed than price changes. The Divisia price-quantity correlations are also mostly negative with an average of −0.36, supporting the *law of demand*.

Figure 2 plots the individual observations (over time) of relative quantity against relative price for each commodity for all countries combined, and Figure 3 plots all the observations in Figure 2 combining all commodities. In the appendix to this paper, available from the corresponding author, we present the mean relative quantities ($Dq_{it} - DQ_t$) and

Table 4. Divisia price and quantity indices, variances and correlations, OECD countries.

	Country (1)	Divisia price index (*DP*) (2)	Divisia quantity index (*DQ*) (3)	Divisia price variance (*Π*) (4)	Divisia quantity variance (*K*) (5)	Divisia price-quantity correlation (ρ) (6)
1	Australia	3.10	1.83	3.01	4.18	−0.42
2	Austria	1.77	0.94	2.27	2.55	−0.40
3	Belgium	1.80	0.73	3.04	3.61	−0.16
4	Canada	2.57	1.52	1.73	3.65	−0.26
5	Colombia	4.53	2.59	2.82	5.22	−0.40
6	Czech Republic	2.85	2.08	7.29	9.43	−0.38
7	Denmark	1.74	0.84	1.79	5.32	−0.46
8	Estonia	5.32	4.63	11.53	25.48	−0.38
9	Finland	3.99	1.76	3.59	7.89	−0.23
10	France	4.16	2.15	2.48	3.66	−0.25
11	Germany	1.24	0.85	1.85	3.72	−0.43
12	Greece	2.59	0.80	4.00	23.29	−0.04
13	Hungary	6.74	1.75	7.34	9.46	−0.22
14	Iceland	4.36	2.20	17.14	31.14	−0.38
15	Ireland	−3.66	7.53	30.42	94.78	−0.88
16	Israel	3.05	1.61	5.66	6.98	−0.14
17	Italy	2.29	0.20	2.16	2.46	−0.17
18	Japan	−0.65	0.85	7.63	13.77	−0.43
19	Korea	7.80	4.20	9.62	17.58	−0.22
20	Latvia	5.24	4.83	29.78	68.43	−0.28
21	Lithuania	3.45	5.76	15.54	43.2	−0.37
22	Luxembourg	2.03	0.66	3.74	11.26	−0.25
23	Mexico	4.42	1.48	6.58	11.44	−0.40
24	Netherlands	2.08	0.72	2.89	15.09	−0.21
25	New Zealand	2.16	1.49	5.35	7.34	−0.42
26	Norway	4.61	2.09	3.71	9.54	−0.38
27	Poland	4.60	3.52	19.62	25.72	−0.38
28	Portugal	1.93	1.64	12.29	12.39	−0.44
29	Slovak Republic	3.49	3.48	10.82	24.27	−0.46
30	Slovenia	4.13	1.76	3.42	7.29	−0.27
31	South Africa	9.45	1.37	12.04	13.82	−0.48
32	Spain	2.52	0.74	2.15	5.62	−0.05
33	Sweden	1.46	1.81	2.62	17.61	−0.35
34	Switzerland	−1.50	2.80	3.10	11.14	−0.92
35	Turkey	16.79	2.03	26.55	29.73	−0.24
36	UK	1.72	1.72	5.91	7.66	−0.68
37	United States	3.60	1.99	3.37	4.31	−0.47
	Mean	3.45	2.13	7.97	16.22	−0.36
	Median	3.05	1.76	4.00	9.54	−0.38
	Std dev	3.26	1.56	7.80	18.7	0.18

Note: Columns (2) and (3) are multiplied by 100. Columns (4) and (5) are multiplied by 10,000.

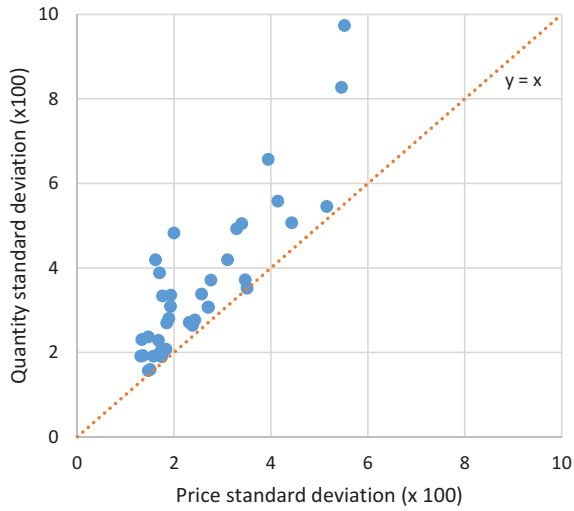

Figure 1. Quantity standard deviation against price standard deviation.

mean relative prices (Dp_{it} – DP_t) for each country and each commodity. It was found that, in a majority of the cases the relative prices and quantities are of the opposite sign, supporting the law of demand that when the relative price of a good increases, its consumption falls. In Figure 4, we plot the mean relative quantities against mean relative prices for each commodity for all countries combined. In Figure 5, we plot all the data points in Figure 4 for all commodities and countries combined. As can be seen from Figure 2–5, the negative slope of all the relative quantity versus relative price graphs again gives support for the law of demand.

[The solid line includes all points and the dotted line omits two outliers (0.59, −1.29) and (−0.16, 4.86).]

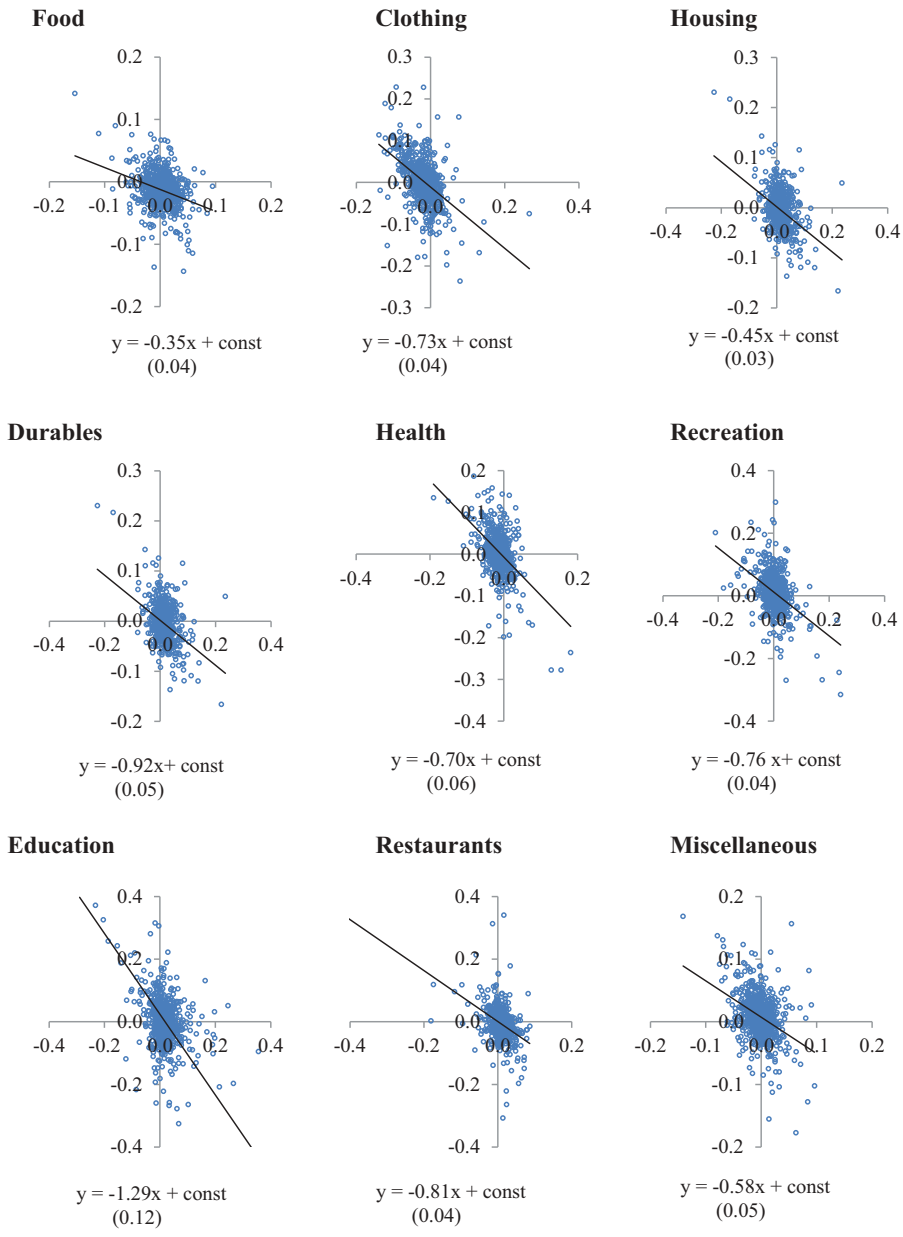

Figure 2. Relative quantity versus relative price, individual observations, all countries combined.

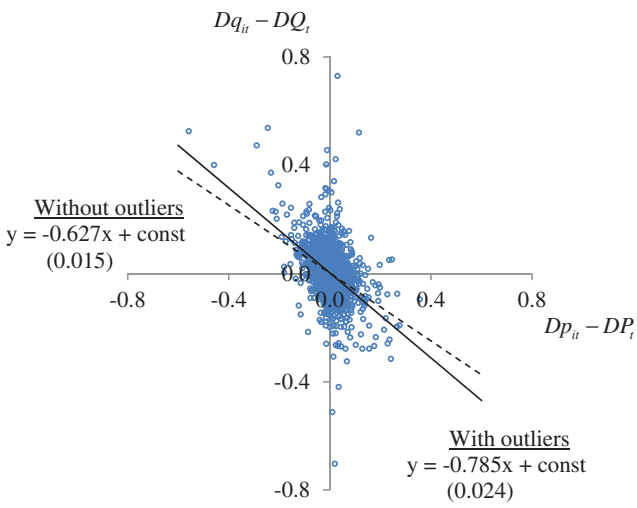

Figure 3. Relative quantity versus relative price, all countries, all commodities combined.

IV. Modelling consumption patterns and estimation results

The Rotterdam model (due to Barten 1964; Theil 1965) is a well-known demand system used widely in empirical applications due to its simplicity and easy interpretation of the parameters (see, Clements and Gao 2015). In this section, we use the Rotterdam model for estimation and derive the implied demand elasticities.

When there are n goods in the consumer's budget, the ith equation of the Rotterdam model is (see, for example, Theil 1980; Selvanathan and Clements 1995; Clements and Gao 2015):

$$\bar{w}_{it} Dq_{it} = \theta_i DQ_t + \sum_{j=1}^{n} \pi_{ij} Dp_{jt} + \varepsilon_{it} \quad i = 1, 2, \ldots, n \tag{1}$$

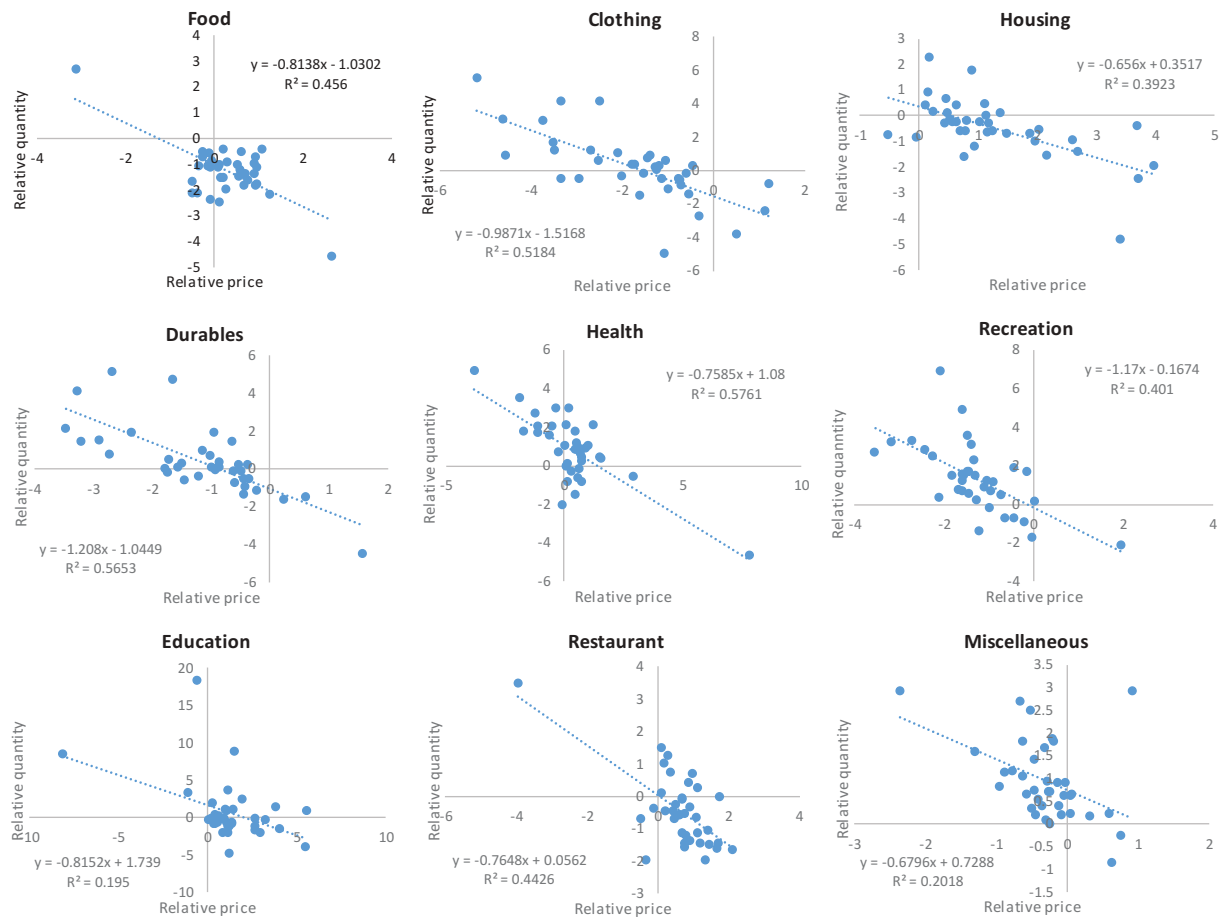

Figure 4. Mean relative quantity versus mean relative price, all countries combined.

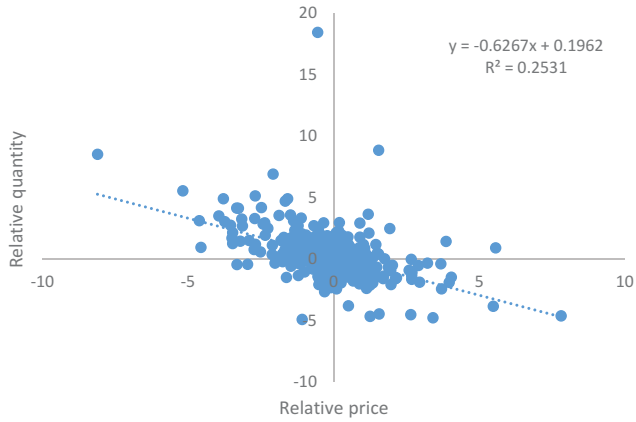

Figure 5. Mean relative quantity versus mean relative price, all countries and all commodities combined.

where $\bar{w}_{it}, Dq_{it}, Dp_{jt}$ and DQ_t are defined as before and $\theta_i = \partial(p_{it}q_{it})/\partial M_t$ is the marginal share of commodity i. The marginal share, θ_i, answers the question that when income increases by one dollar, how much of this increase is spent on commodity i? The coefficient π_{ij} is the (i,j)th Slutsky price coefficient which satisfies the adding-up restriction, $\sum_{i=1}^{n} \pi_{ij} = 0$. These price coefficients also satisfy the following constraints:

$$\sum_{j=1}^{n} \pi_{ij} = 0 \qquad i = 1, 2, \ldots, n \tag{2}$$

Constraint (2) reflects the homogeneity property of the demand system that postulates that an equi-proportionate change in all prices has no effect on the demand for any good under the condition that real income is held constant. Accordingly, equation (2) is known as *demand homogeneity*.

The Slutsky coefficients are symmetric in i and j, that is

Table 5. Hypothesis test results.

		Testing Homogeneity			Testing Symmetry given Homogeneity		
	Country (1)	Test statistic (2)	Critical value, $F_{(0.05,n-1,T-2n+1)}$ (3)	Decision (4)	Test statistic (5)	Critical value, $\chi^2_{(0.05,q)}$ (6)	Decision (7)
1	Australia	0.54	2.85	Accept	45.13	41.34	Reject
2	Austria	9.13	19.40	Accept	245.48	41.34	Reject
3	Belgium	4.61	19.40	Accept	40.32	41.34	Accept
4	Canada	4.62	2.55	Reject	102.23	41.34	Reject
5	Colombia[a]						
6	Czech Republic	0.45	8.85	Accept	68.48	41.34	Reject
7	Denmark	0.50	8.85	Accept	141.26	41.34	Reject
8	Estonia	0.59	8.85	Accept	150.06	41.34	Reject
9	Finland	4.06	2.38	Reject	78.24	41.34	Reject
10	France	2.87	2.18	Reject	52.48	41.34	Reject
11	Germany	0.43	19.40	Accept	102.55	41.34	Reject
12	Greece	0.26	8.85	Accept	53.15	41.34	Reject
13	Hungary	3.28	8.85	Accept	99.22	41.34	Reject
14	Iceland	7.89	19.40	Accept	163.11	41.34	Reject
15	Ireland	14.25	239.00	Accept	236.89	41.34	Reject
16	Israel	0.95	19.40	Accept	161.47	41.34	Reject
17	Italy	3.05	19.40	Accept	96.79	41.34	Reject
18	Japan	0.40	19.40	Accept	144.54	41.34	Reject
19	Korea	1.99	2.29	Accept	46.82	41.34	Reject
20	Latvia	2.32	19.40	Accept	54.11	41.34	Reject
21	Lithuania	2.14	19.40	Accept	48.92	41.34	Reject
22	Luxembourg	0.33	8.85	Accept	77.48	41.34	Reject
23	Mexico[a]						
24	Netherlands	17.97	19.40	Accept	39.03	41.34	Accept
25	New Zealand	1.46	3.07	Accept	47.9	41.34	Reject
26	Norway	3.89	2.30	Reject	59.1	41.34	Reject
27	Poland	2.07	8.85	Accept	100.62	41.34	Reject
28	Portugal	2.08	8.85	Accept	61.31	41.34	Reject
29	Slovak Republic	6.10	8.85	Accept	63.11	41.34	Reject
30	Slovenia	2.77	8.85	Accept	80.83	41.34	Reject
31	South Africa	2.43	2.40	Reject	57.47	41.34	Reject
32	Spain	6.50	19.40	Accept	460.16	41.34	Reject
33	Sweden	2.36	4.82	Accept	27.44	41.34	Accept
34	Switzerland	0.52	239.00	Accept	55.42	41.34	Reject
35	Turkey[a]						
36	UK	0.47	8.85	Accept	66.59	41.34	Reject
37	United States	3.95	2.19	Reject	81.06	41.34	Reject

Note: [a] Due to data limitation, the tests could not be carried out for Columbia, Mexico and Turkey.

Table 6. Preference independence hypothesis test results.

	Country (1)	Simulated test statistic Low (2)	High (3)	Observed value of the test statistic (4)	Rank (5)	Decision (6)
1	Australia	19.02	120.33	69.55	891.00	Accept
2	Austria	22.72	162.87	71.71	532.00	Accept
3	Belgium	22.94	162.66	95.70	762.00	Accept
4	Canada	12.66	95.17	108.84	1000.00	Reject
5	Colombia[b]					
6	Czech Republic	19.49	163.93	70.20	508.00	Accept
7	Denmark	23.00	164.91	59.74	255.00	Accept
8	Estonia	22.81	186.95	72.98	501.00	Accept
9	Finland	14.27	93.45	51.92	728.00	Accept
10	France	4.15	99.25	92.98	999.00	Reject
11	Germany	30.14	198.94	56.27	156.00	Accept
12	Greece	26.69	159.98	95.89	860.00	Accept
13	Hungary	18.79	176.69	61.15	302.00	Accept
14	Iceland	21.46	208.51	83.65	660.00	Accept
15	Ireland	53.73	216.08	146.33	997.00	Reject
16	Israel	23.67	184.37	61.69	202.00	Accept
17	Italy	25.59	167.24	73.47	449.00	Accept
18	Japan	19.08	210.34	96.67	830.00	Accept
19	Korea	13.20	83.31	62.61	954.00	Accept[a]
20	Latvia	13.70	187.84	122.58	955.00	Accept[a]
21	Lithuania	26.46	167.65	113.54	931.00	Accept
22	Luxembourg	24.75	164.27	85.61	806.00	Accept
23	Mexico[b]					
24	Netherlands	28.48	192.68	11.09	882.00	Accept
25	New Zealand	16.01	118.86	91.52	993.00	Reject
26	Norway	1.91	88.67	47.57	805.00	Accept
27	Poland	13.48	177.19	92.98	849.00	Accept
28	Portugal	21.33	137.08	107.98	938.00	Accept
29	Slovak Republic	23.02	200.44	113.93	973.00	Accept[a]
30	Slovenia	19.41	144.73	91.97	833.00	Accept
31	South Africa	12.96	90.09	66.64	965.00	Accept[a]
32	Spain	32.89	172.70	68.26	334.00	Accept
33	Sweden	17.61	131.79	96.18	953.00	Accept[a]
34	Switzerland	1.69	226.53	112.23	816.00	Accept
35	Turkey	8.47	99.24	99.86	1000.00	Reject
36	UK	13.71	160.23	118.51	975.00	Accept[a]
37	United States	8.64	88.18	109.27	1000.00	Reject

[a] Conclusion significant at the 10% level.
[b] Due to data limitation, the test of preference independence could not be carried out for Mexico and Colombia due to near singularity.

$$\pi_{ij} = \pi_{ji} \qquad i, j = 1, 2, \ldots, n \tag{3}$$

which is known as *Slutsky symmetry*. This symmetry property means that an increase in the price of any good j will cause an increase in the compensated quantity demanded of i equal to the increase in the compensated quantity demanded of j caused by an increase in the price of i. In other words, as the commodity subscripts can be interchanged, the substitution effects are symmetric in i and j. Also, the Slutsky matrix$[\pi_{ij}]$ is symmetric and negative semi-definite with rank (n-1).

The income and price elasticities implied by demand system (1) are given by

$$\eta_{it} = \frac{\theta_i}{\bar{w}_{it}} \quad and \quad \eta_{ijt} = \frac{\pi_{ij}}{\bar{w}_{it}}.$$

If the income elasticity of good i is less (greater) than 1, then good i is classified as a necessity (luxury). To calculate these elasticities, we use the sample means of the budget shares

$$\bar{w}_i = \frac{1}{T}\sum_{t=1}^{T} \bar{w}_{it}$$

and calculate

$$\bar{\eta}_i = \frac{\theta_i}{\bar{w}_i} \quad and \quad \bar{\eta}_{ij} = \frac{\pi_{ij}}{\bar{w}_i}. \tag{4}$$

We now test the above two demand theory hypotheses demand homogeneity (2) and Slutsky symmetry (3) to see whether OECD consumers support these two hypotheses. The standard test for homogeneity hypothesis has an asymptotic χ^2 distribution with (n-1) degrees of freedom. Due to

Table 7. Estimates of demand equations under preference independence $\bar{w}_{it}(Dq_{it} - DQ_t) = \beta_i DQ_t + \phi(\bar{w}_i + \beta_i)\left(Dp_{it} - \Sigma_{j=1}^{n}(\bar{w}_j + \beta_j)Dp_{jt}\right) + \varepsilon_{it}$.

	Income coefficient β_i									
Country (1)	Food (2)	Clothing (3)	Housing (4)	Durables (5)	Health (6)	Recreation (7)	Education (8)	Restaurants (9)	Miscellaneous (10)	Income flexibility ϕ (11)
1. Australia	−0.096 (0.016)	−0.007 (0.007)	−0.047 (0.014)	−0.011 (0.008)	0.012 (0.011)	0.036 (0.013)	0.002 (0.006)	0.012 (0.016)	0.098 (0.029)	−0.414 (0.054)
2. Austria	−0.099 (0.015)	−0.024 (0.012)	−0.024 (0.042)	−0.002 (0.009)	−0.007 (0.010)	0.055 (0.017)	0.002 (0.002)	−0.032 (0.030)	0.131 (0.039)	−0.463 (0.062)
3. Belgium	−0.216 (0.041)	−0.068 (0.027)	−0.056 (0.043)	0.027 (0.019)	0.010 (0.014)	0.061 (0.046)	−0.002 (0.002)	−0.003 (0.017)	0.248 (0.061)	−0.050 (0.047)
4. Canada	−0.112 (0.014)	0.014 (0.007)	−0.141 (0.023)	0.038 (0.008)	0.005 (0.006)	0.038 (0.010)	−0.003 (0.001)	0.016 (0.010)	0.144 (0.022)	−0.350 (0.048)
5. Colombia	−0.137 (0.020)	0.019 (0.015)	−0.073 (0.012)	0.002 (0.005)	−0.014 (0.007)	0.043 (0.011)	−0.032 (0.007)	0.016 (0.014)	0.176 (0.025)	−0.082 (0.045)
6. Czech Rep.	−0.098 (0.026)	0.003 (0.006)	−0.145 (0.032)	0.015 (0.008)	0.006 (0.008)	0.029 (0.014)	0.004 (0.003)	−0.007 (0.020)	0.192 (0.030)	−0.438 (0.067)
7. Denmark	−0.107 (0.011)	−0.015 (0.012)	−0.223 (0.043)	0.023 (0.010)	−0.012 (0.007)	0.059 (0.020)	−0.008 (0.003)	0.023 (0.015)	0.259 (0.031)	−0.530 (0.087)
8. Estonia	−0.110 (0.027)	0.006 (0.008)	−0.040 (0.027)	0.019 (0.006)	0.000 (0.003)	0.045 (0.009)	−0.001 (0.002)	0.004 (0.009)	0.077 (0.019)	−0.321 (0.055)
9. Finland	−0.130 (0.014)	0.018 (0.008)	−0.141 (0.022)	0.025 (0.007)	0.002 (0.006)	0.049 (0.010)	−0.001 (0.002)	0.015 (0.008)	0.164 (0.023)	−0.486 (0.059)
10. France	−0.123 (0.006)	−0.031 (0.007)	0.034 (0.019)	0.004 (0.005)	0.006 (0.004)	0.034 (0.005)	−0.001 (0.002)	−0.025 (0.006)	0.101 (0.016)	−0.444 (0.046)
11. Germany	−0.086 (0.035)	−0.018 (0.015)	−0.147 (0.061)	0.031 (0.036)	−0.005 (0.021)	0.072 (0.026)	0.011 (0.003)	0.027 (0.019)	0.114 (0.054)	−0.300 (0.051)
12. Greece	−0.057 (0.043)	0.034 (0.014)	−0.123 (0.031)	0.040 (0.012)	0.039 (0.018)	0.020 (0.010)	0.002 (0.008)	−0.028 (0.033)	0.072 (0.055)	−0.148 (0.134)
13. Hungary	−0.098 (0.018)	−0.002 (0.006)	−0.068 (0.019)	0.013 (0.012)	0.036 (0.009)	0.012 (0.008)	−0.008 (0.006)	−0.010 (0.015)	0.127 (0.030)	−0.524 (0.052)
14. Iceland	−0.115 (0.016)	0.000 (0.008)	0.040 (0.039)	−0.008 (0.011)	−0.019 (0.005)	−0.042 (0.014)	−0.012 (0.001)	0.010 (0.016)	0.146 (0.031)	−0.806 (0.095)
15. Ireland	−0.046 (0.007)	−0.006 (0.001)	0.163 (0.012)	−0.016 (0.001)	−0.016 (0.002)	−0.013 (0.001)	−0.005 (0.001)	−0.019 (0.003)	−0.042 (0.007)	−2.459 (0.053)
16. Israel	−0.116 (0.017)	−0.002 (0.015)	−0.107 (0.023)	0.034 (0.014)	0.005 (0.006)	0.022 (0.015)	−0.004 (0.007)	−0.005 (0.036)	0.173 (0.028)	−0.017 (0.059)
17. Italy	−0.052 (0.013)	0.021 (0.012)	−0.171 (0.022)	0.032 (0.013)	0.000 (0.007)	0.060 (0.012)	−0.001 (0.001)	0.008 (0.020)	0.102 (0.028)	−0.526 (0.048)
18. Japan	−0.090 (0.047)	−0.014 (0.034)	−0.143 (0.053)	0.014 (0.015)	−0.056 (0.018)	0.064 (0.045)	−0.026 (0.006)	−0.024 (0.023)	0.277 (0.085)	−0.605 (0.160)
19. Korea	−0.120 (0.014)	0.021 (0.011)	−0.050 (0.014)	0.028 (0.006)	0.011 (0.007)	0.046 (0.005)	−0.022 (0.006)	−0.007 (0.008)	0.091 (0.012)	−0.209 (0.046)
20. Latvia	−0.104 (0.029)	0.031 (0.010)	−0.160 (0.038)	0.027 (0.005)	0.004 (0.006)	0.052 (0.017)	0.004 (0.009)	0.046 (0.023)	0.100 (0.026)	−0.423 (0.106)
21. Lithuania	−0.089 (0.023)	0.032 (0.012)	−0.113 (0.020)	0.018 (0.009)	0.014 (0.011)	0.035 (0.011)	0.000 (0.002)	0.005 (0.007)	0.099 (0.031)	−0.394 (0.089)
22. Luxembourg	0.208 (0.049)	−0.010 (0.012)	−0.031 (0.040)	−0.009 (0.013)	−0.008 (0.007)	−0.043 (0.014)	−0.003 (0.003)	0.000 (0.013)	−0.106 (0.048)	−0.848 (0.143)
23. Mexico	−0.109 (0.000)	−0.025 (0.000)	0.016 (0.000)	−0.017 (0.000)	−0.019 (0.000)	0.055 (0.000)	−0.016 (0.000)	−0.016 (0.000)	0.131 (0.000)	−0.830 (0.000)
24. Netherlands	−0.083 (0.010)	0.004 (0.011)	−0.166 (0.032)	0.032 (0.016)	0.030 (0.068)	0.043 (0.020)	−0.004 (0.003)	0.013 (0.016)	0.131 (0.040)	−0.282 (0.066)
25. New Zealand	−0.047 (0.018)	0.007 (0.007)	−0.156 (0.019)	0.021 (0.012)	−0.021 (0.008)	0.015 (0.014)	−0.003 (0.002)	0.041 (0.016)	0.142 (0.026)	−0.624 (0.067)
26. Norway	−0.076 (0.014)	−0.001 (0.005)	−0.137 (0.015)	0.006 (0.007)	−0.010 (0.004)	0.022 (0.010)	−0.003 (0.002)	−0.007 (0.008)	0.206 (0.025)	−0.906 (0.082)
27. Poland	−0.173 (0.017)	0.008 (0.008)	−0.064 (0.038)	0.020 (0.008)	0.037 (0.008)	0.027 (0.018)	−0.003 (0.003)	0.020 (0.006)	0.128 (0.032)	−0.606 (0.071)
28. Portugal	0.050 (0.030)	0.013 (0.011)	−0.090 (0.014)	−0.031 (0.007)	0.010 (0.010)	−0.049 (0.010)	−0.001 (0.004)	−0.007 (0.015)	0.104 (0.051)	−0.612 (0.083)
29. Slovakia	−0.120 (0.025)	−0.027 (0.012)	−0.007 (0.041)	0.028 (0.009)	0.006 (0.012)	0.038 (0.010)	0.005 (0.003)	0.012 (0.013)	0.066 (0.019)	−0.477 (0.058)
30. Slovenia	−0.072 (0.022)	0.032 (0.013)	−0.076 (0.025)	0.029 (0.017)	0.009 (0.011)	0.043 (0.023)	0.001 (0.004)	−0.007 (0.013)	0.041 (0.040)	−0.336 (0.084)
31. South Africa	−0.121 (0.030)	0.021 (0.008)	−0.062 (0.020)	0.004 (0.009)	−0.001 (0.012)	0.009 (0.008)	−0.006 (0.004)	0.013 (0.008)	0.144 (0.027)	−0.376 (0.046)
32. Spain	−0.106 (0.023)	0.016 (0.012)	−0.106 (0.044)	0.017 (0.009)	−0.008 (0.008)	0.036 (0.010)	−0.014 (0.003)	0.038 (0.031)	0.126 (0.024)	−0.219 (0.062)
33. Sweden	−0.081 (0.020)	0.027 (0.008)	−0.217 (0.020)	0.044 (0.008)	−0.024 (0.011)	0.061 (0.013)	0.038 (0.006)	−0.016 (0.012)	0.168 (0.043)	−0.515 (0.068)
34. Switzerland	0.047 (0.006)	−0.002 (0.002)	0.007 (0.007)	−0.001 (0.001)	−0.003 (0.003)	−0.007 (0.002)	0.000 (0.000)	0.009 (0.002)	−0.052 (0.007)	−1.706 (0.031)
35. Turkey	−0.179 (0.030)	−0.039 (0.031)	−0.095 (0.017)	0.062 (0.016)	0.011 (0.011)	0.028 (0.013)	0.004 (0.003)	−0.025 (0.025)	0.233 (0.035)	−0.183 (0.039)
36. UK	−0.070 (0.007)	0.008 (0.006)	−0.146 (0.018)	0.014 (0.008)	0.001 (0.006)	0.041 (0.013)	−0.006 (0.002)	0.012 (0.015)	0.146 (0.038)	−0.812 (0.078)
37. United States	−0.085 (0.007)	0.000 (0.004)	−0.073 (0.011)	0.016 (0.006)	−0.033 (0.019)	0.041 (0.006)	−0.006 (0.002)	−0.009 (0.006)	0.148 (0.021)	−0.601 (0.047)
Unweighted mean	−0.087 (0.020)	0.001 (0.011)	−0.085 (0.026)	0.016 (0.010)	0.000 (0.010)	0.031 (0.014)	−0.003 (0.003)	0.003 (0.015)	0.124 (0.031)	−0.538 (0.067)
Weighted mean	−0.108 (0.000)	−0.024 (0.000)	0.016 (0.000)	−0.016 (0.000)	−0.019 (0.000)	0.054 (0.000)	−0.016 (0.000)	−0.016 (0.000)	0.131 (0.000)	−0.828 (0.000)
Median	−0.098	0.003	−0.090	0.018	0.001	0.038	−0.003	0.004	0.131	−0.463
SD	0.069	0.022	0.077	0.019	0.019	0.029	0.011	0.019	0.078	0.438

Note: Miscellaneous goods and services include transport and communication. Weights for each parameter are $\left[1/SE(\beta_i^c)\right]/\Sigma_{c=1}^{C}\left[1/SE(\beta_i^c)\right]$. Standard errors are in parentheses.

Table 8. Income elasticities of demand, 9 commodities, 37 countries.

Country (1)	Food (2)	Clothing (3)	Housing (4)	Durables (5)	Health (6)	Recreation (7)	Education (8)	Restaurants (9)	Miscellaneous (10)
1. Australia	0.34, 0.29	0.83, 0.77	0.77, 0.80	0.80, 0.74	1.24, 1.19	1.33, 1.37	1.07, 1.05	1.18, 1.19	1.34, 1.35
2. Austria	0.28, 0.26	0.62, 0.60	0.88, 0.89	0.97, 0.97	0.82, 0.82	1.53, 1.54	1.22, 1.21	0.72, 0.75	1.51, 1.54
3. Belgium	−0.27, −0.28*	−0.36, −0.47*	0.77, 0.77	1.44, 1.43	1.18, 1.16	1.66, 1.70	0.52, 0.50	0.94, 0.94	1.91, 1.94
4. Canada	0.23, 0.12	1.26, 1.34	0.41, 0.43	1.63, 1.71	1.14, 1.11	1.43, 1.46	0.75, 0.83	1.22, 1.24	1.48, 1.45
5. Colombia	0.38, 0.34	1.29, 1.30	0.56, 0.54	1.04, 1.04	0.57, 0.53	1.87, 1.82	0.36, 0.34	1.14, 1.13	1.68, 1.64
6. Czech Rep.	0.59, 0.61	1.07, 1.09	0.41, 0.44	1.27, 1.28	1.30, 1.26	1.29, 1.34	1.75, 1.82	0.91, 0.92	1.92, 1.94
7. Denmark	0.33, 0.29	0.67, 0.65	0.18, 0.24	1.43, 1.46	0.56, 0.59	1.53, 1.55	−0.05, 0.01*	1.43, 1.39	1.97, 2.00
8. Estonia	0.61, 0.63	1.09, 1.09	0.80, 0.78	1.42, 1.46	0.99, 1.00	1.55, 1.56	0.87, 0.74	1.06, 1.05	1.35, 1.35
9. Finland	0.40, 0.24	1.36, 1.42	0.37, 0.49	1.48, 1.50	1.05, 1.04	1.45, 1.46	0.67, 0.70	1.22, 1.24	1.68, 1.69
10. France	0.40, 0.27	0.58, 0.21	1.17, 1.13	1.07, 1.09	1.19, 1.13	1.39, 1.41	0.92, 0.94	0.59, 0.63	1.38, 1.37
11. Germany	0.40, 0.36	0.67, 0.64	0.39, 0.40	1.44, 1.46	0.89, 0.91	1.75, 1.76	2.57, 2.36*	1.54, 1.52	1.39, 1.39
12. Greece	0.73, 0.74	1.68, 1.99	0.35, 0.41	1.91, 2.24	1.87, 1.94	1.43, 1.43	1.12, 1.11	0.80, 0.81	1.28, 1.28
13. Hungary	0.61, 0.63	0.94, 0.94	0.65, 0.64	1.23, 1.29	1.90, 1.77	1.15, 1.17	0.44, 0.48	0.84, 0.87	1.48, 1.49
14. Iceland	0.40, 0.39	0.99, 0.99	1.20, 1.18	0.87, 0.88	0.38, 0.37	0.63, 0.60	0.07, 0.09	1.12, 1.10	1.57, 1.60
15. Ireland	0.73, 0.71	0.87, 0.83	1.81, 1.69*	0.74, 0.62	0.55, 0.67	0.81, 0.79	0.79, 0.86	0.86, 0.87	0.83, 0.82
16. Israel	0.36, 0.40	0.96, 0.94	0.54, 0.57	1.50, 1.60	1.16, 1.17	1.31, 1.38	0.86, 0.86	0.91, 0.92	1.58, 1.59
17. Italy	0.73, 0.72	1.31, 1.35	0.17, 0.30	1.46, 1.53	1.00, 1.00	1.85, 1.92	0.86, 0.87	1.09, 1.09	1.39, 1.42
18. Japan	0.48, 0.46	0.67, 0.60	0.40, 0.43	1.32, 1.34	−0.46, −0.22*	1.61, 1.71	−0.19, −0.22*	0.64, 0.62	2.03, 1.98
19. Korea	0.55, 0.24	1.28, 1.35	0.66, 0.73	1.70, 1.97	1.31, 1.23	1.72, 1.55	0.60, 0.62	0.91, 0.91	1.39, 1.31
20. Latvia	0.68, 0.63	1.60, 1.60	0.21, 0.27	1.89, 1.81	1.09, 1.09	1.62, 1.61	1.23, 1.20	1.83, 1.78	1.51, 1.48
21. Lithuania	0.74, 0.71	1.50, 1.52	0.30, 0.31	1.33, 1.29	1.33, 1.29	1.55, 1.48	0.94, 0.93	1.15, 1.16	1.43, 1.39
22. Luxembourg	2.11, 2.18*	0.82, 0.81	0.86, 0.87	0.87, 0.84	0.60, 0.65	0.40, 0.33	0.62, 0.76	1.01, 1.01	0.66, 0.67
23. Mexico	0.57, 0.58	0.17, 0.20	1.07, 1.08	0.71, 0.70	0.56, 0.51	2.29, 2.19	−0.06, −0.09*	0.63, 0.60	1.43, 1.42
24. Netherlands	0.42, 0.45	1.08, 1.09	0.18, 0.30	1.50, 1.61	1.77, 1.83	1.38, 1.39	0.38, 0.44	1.18, 1.18	1.43, 1.47
25. New Zealand	0.76, 0.75	1.17, 1.19	0.39, 0.42	1.34, 1.39	0.06, 0.15	1.15, 1.17	0.72, 0.82	1.71, 1.66	1.56, 1.55
26. Norway	0.64, 0.54	0.98, 0.97	0.36, 0.37	1.09, 1.10	0.56, 0.65	1.21, 1.20	0.31, 0.40	0.86, 0.89	1.81, 1.71
27. Poland	0.40, 0.28	1.16, 1.16	0.69, 0.71	1.45, 1.39	1.93, 1.75	1.35, 1.35	0.72, 0.68	1.67, 1.67	1.51, 1.46
28. Portugal	1.25, 1.25*	1.21, 1.21	0.40, 0.54	0.50, 0.39	1.20, 1.19	0.33, 0.15	0.95, 0.95	0.93, 0.94	1.37, 1.40
29. Slovakia	0.53, 0.48	0.42, 0.31	0.97, 0.97	1.50, 1.47	1.20, 1.25	1.43, 1.39	1.45, 1.32	1.18, 1.21	1.32, 1.29
30. Slovenia	0.66, 0.66	1.57, 1.63	0.60, 0.60	1.51, 1.58	1.25, 1.22	1.44, 1.50	1.06, 1.05	0.90, 0.91	1.15, 1.14
31. South Africa	0.58, 0.52	1.33, 1.43	0.52, 0.58	1.04, 1.07	0.98, 0.99	1.15, 1.20	0.73, 0.83	1.46, 1.49	1.54, 1.46
32. Spain	0.41, 0.39	1.30, 1.38	0.43, 0.56	1.32, 1.39	0.77, 0.81	1.44, 1.52	0.13, 0.25	1.23, 1.27	1.54, 1.55
33. Sweden	0.51, 0.50	1.56, 1.56	0.22, 0.17	1.93, 1.84	0.21, 0.32	1.55, 1.56	24.00, 14.52*	0.69, 0.74	1.63, 1.62
34. Switzerland	1.35, 1.38*	0.96, 0.95	1.03, 1.03	0.98, 0.98	0.98, 0.98	0.93, 0.92	1.07, 1.06	1.13, 1.13	0.78, 0.78
35. Turkey	0.33, 0.30	0.44, 0.18	0.47, 0.48	1.81, 1.83	1.30, 1.35	1.63, 1.73	1.37, 1.31	0.60, 0.64	1.91, 1.80
36. UK	0.45, 0.44	1.16, 1.15	0.42, 0.43	1.28, 1.30	1.04, 1.04	1.39, 1.43	0.53, 0.61	1.13, 1.13	1.51, 1.51
37. United States	0.27, 0.00	0.99, 0.99	0.61, 0.61	1.30, 1.38	0.79, 0.85	1.48, 1.45	0.66, 0.74	0.87, 0.87	1.56, 1.57
Mean	0.57, 0.52	1.01, 1.00	0.60, 0.63	1.30, 1.32	0.98, 0.99	1.38, 1.38	1.41, 1.16	1.06, 1.07	1.48, 1.47
Median	0.51, 0.46	1.07, 1.09	0.52, 0.56	1.33, 1.39	1.04, 1.04	1.43, 1.45	0.75, 0.83	1.06, 1.05	1.51, 1.47
SD	0.37, 0.40	0.42, 0.48	0.34, 0.31	0.34, 0.39	0.49, 0.45	0.38, 0.40	3.80, 2.28	0.31, 0.30	0.29, 0.29

Note: The income elasticity of good i in country c is $1 + \beta_i^c / w_{ic}$, where for commodity i and country c, β_i^c is the income coefficient and w_{ic} is the budget share. We use the β_i^c estimates from Table 7. The first entry in each cell is the income elasticity evaluated with the sample mean of w_{ic}, the second entry uses w_{ic} in the last year of the sample. Miscellaneous goods and services include transport and communication. An asterisk refers to some of the 'odd' cases and outliers identified by the box plots.

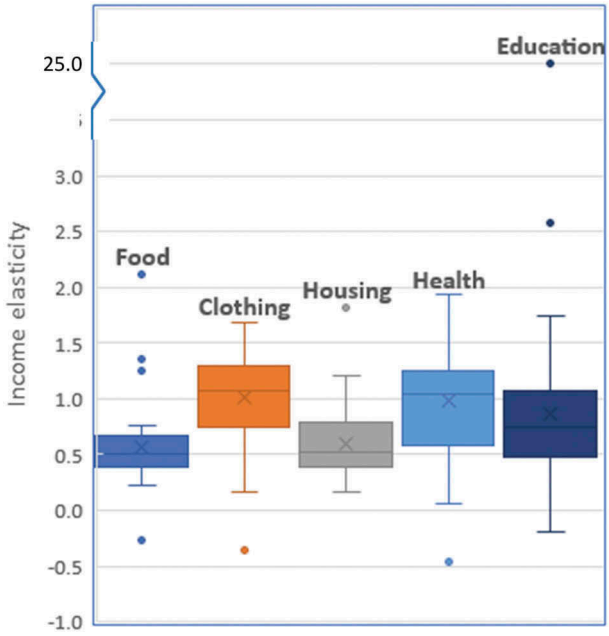

Figure 6. Income elasticities, 37 countries, 5 commodities.

the small sample sizes for many countries in our data set, we use the exact finite sample test derived by Laitinen (1978). This test involves the error covariance matrix and the necessary condition for the error covariance matrix to be non-singular is that T–2n–1 is non-negative (Laitinen, 1978), where n is the number of commodities and T is the number of observations.

The test statistic to test symmetry given homogeneity has an asymptotic χ^2 distribution with q = ½(n-1)(n-2) degrees of freedom. Using simulation experiments, Meisner (1979) showed that the asymptotic test is biased against symmetry, particularly in large demand systems. Since the symmetry constraint (3) involves cross-equation restrictions, the exact distribution of the test statistic is complicated and, to-date, has not been derived.

Table 5 presents the test results for homogeneity and symmetry. As can be seen, the homogeneity hypothesis is supported about 90 per cent the OECD countries consumption data while symmetry hypothesis is mostly rejected. As stated above, the reason for rejection for symmetry could be due to the use of the asymptotic test while the sample size we have relatively small (Meisner 1979). Further investigation is needed with the symmetry test results when more data becomes available.

When a consumer's utility function can be expressed as the sum of n sub-utility functions, one for each commodity, such condition is called *preference independence*. Preference independence also implies that the marginal utility of commodity i is independent of another commodity j, for i ≠ j. The 9 broad commodity classifications in this study can conceivably be interpreted as representing the basic wants of the consumer, which in turn, could be expected to exhibit scant interaction in the utility function. Below we test the preference independent structure for individual countries and then consider the demand system under preference independence for estimation.

Under preference independence, the Slutsky coefficients take the form (see, for example, Selvanathan and Selvanathan 1994):

$$\pi_{ij} = \phi\theta_i(\delta_{ij} - \theta_j) \tag{5}$$

where ϕ is the income flexibility (the reciprocal of the income elasticity of the marginal utility of income) and δ_{ij} is the Kronecker delta. Substituting (5) for π_{ij} in (1), the preference independence version of (1) is

$$\bar{w}_{it}Dq_{it} = \theta_i DQ_t + \phi\theta_i\left(Dp_{it} - \Sigma_{j=1}^{n}\theta_j Dp_{jt}\right) + \varepsilon_{it} \quad i = 1, 2, \ldots, n. \tag{6}$$

The demand equation here is much simplified as it contains only the own-relative price. Another attraction of preference independence is that the number of parameters to be estimated is significantly reduced. For example, when n = 9, to estimate demand system (1), we need to estimate 8 free marginal shares (θ_i, i = 1,2, . . ., 8) and 8 × 9 = 72 Slutsky coefficients (π_{ij}, i = 1,2, . . ., 8, j = 1,2, . . ., 9), giving a total of 80 free parameters. Under preference independence, in model (6), we need to estimate only 8 free marginal shares (θ_i, i = 1,2, . . ., 8) and 1 income flexibility (ϕ), giving a total 9 free parameters to be estimated. As we have shorter time series data, the preference independence version of the model is probably a more suitable approach to use.

To avoid problems with the assumption that the marginal shares are constants, we use Working's (1943) model. This model implies that the marginal share of i exceeds the corresponding budget share by a constant, β_i. In a time-series context, this amounts to $\theta_{it} = \bar{w}_{it} + \beta_i$. Thus, substituting $\bar{w}_{it} + \beta_i$ for θ_i

Table 9. Compensated own-price elasticities of demand, 9 commodities, 37 countries.

Country (1)	Food (2)	Clothing (3)	Housing (4)	Durables (5)	Health (6)	Recreation (7)	Education (8)	Restaurants (9)	Miscellaneous (10)
1. Australia	−0.13, −0.11	−0.33, −0.31	−0.27, −0.27	−0.32, −0.30	−0.48, −0.46	−0.47, −0.49	−0.43, −0.41	−0.45, −0.45	−0.34, −0.35
2. Austria	−0.12, −0.12	−0.28, −0.27	−0.33, −0.33	−0.42, −0.42	−0.37, −0.37	−0.59, −0.60	−0.56, −0.55	−0.31, −0.31	−0.43, −0.45
3. Belgium	-	-	−0.03, −0.03	−0.07, −0.07	−0.05, −0.05	−0.07, −0.07	−0.03, −0.02	−0.04, −0.04	−0.05, −0.05
4. Canada	−0.08, −0.04	−0.41, −0.44	−0.13, −0.13	−0.51, −0.54	−0.38, −0.37	−0.43, −0.45	−0.26, −0.29	−0.39, −0.40	−0.29, −0.27
5. Colombia	−0.03, −0.03	−0.10, −0.10	−0.04, −0.04	−0.08, −0.08	−0.05, −0.04	−0.14, −0.13	−0.03, −0.03	−0.08, −0.08	−0.08, −0.07
6. Czech Rep.	−0.22, −0.23	−0.45, −0.46	−0.16, −0.17	−0.52, −0.52	−0.56, −0.54	−0.49, −0.52	−0.76, −0.79	−0.37, −0.37	−0.50, −0.51
7. Denmark	−0.16, −0.15	−0.35, −0.33	−0.09, −0.12	−0.70, −0.72	−0.29, −0.31	−0.67, −0.69	-	−0.70, −0.67	−0.49, −0.51
8. Estonia	−0.16, −0.16	−0.33, −0.32	−0.22, −0.21	−0.42, −0.44	−0.31, −0.31	−0.43, −0.44	−0.28, −0.24	−0.32, −0.31	−0.30, −0.30
9. Finland	−0.18, −0.11	−0.61, −0.65	−0.17, −0.21	−0.66, −0.68	−0.49, −0.48	−0.59, −0.60	−0.33, −0.34	−0.55, −0.55	−0.49, −0.49
10. France	−0.16, −0.11	−0.25, −0.09	−0.39, −0.35	−0.44, −0.46	−0.51, −0.48	−0.54, −0.55	−0.41, −0.42	−0.25, −0.27	−0.39, −0.38
11. Germany	−0.11, −0.10	−0.19, −0.18	−0.11, −0.11	−0.39, −0.39	−0.26, −0.26	−0.44, −0.44	-	−0.43, −0.42	−0.25, −0.24
12. Greece	−0.09, −0.09	−0.23, −0.27	−0.05, −0.06	−0.26, −0.31	−0.25, −0.26	−0.20, −0.20	−0.16, −0.16	−0.10, −0.11	−0.13, −0.13
13. Hungary	−0.27, −0.27	−0.48, −0.48	−0.30, −0.30	−0.60, −0.64	−0.92, −0.85	−0.55, −0.56	−0.23, −0.25	−0.42, −0.42	−0.47, −0.48
14. Iceland	−0.30, −0.29	−0.76, −0.77	−0.74, −0.71	−0.66, −0.66	−0.30, −0.29	−0.47, −0.45	−0.05, −0.07	−0.82, −0.79	−0.76, −0.78
15. Ireland	−1.57, −1.55	−2.06, −1.98	-	−1.74, −1.49	−1.33, −1.59	−1.88, −1.85	−1.91, −2.04	−1.86, −1.86	−1.62, −1.63
16. Israel	−0.01, −0.01	−0.02, −0.02	−0.01, −0.01	−0.02, −0.02	−0.02, −0.02	−0.02, −0.02	−0.01, −0.01	−0.01, −0.01	−0.01, −0.01
17. Italy	−0.33, −0.33	−0.62, −0.65	−0.09, −0.15	−0.69, −0.73	−0.51, −0.51	−0.85, −0.88	−0.45, −0.45	−0.52, −0.51	−0.47, −0.49
18. Japan	−0.27, −0.26	−0.40, −0.35	−0.22, −0.23	−0.75, −0.77	-	−0.81, −0.87	-	−0.37, −0.36	−0.56, −0.53
19. Korea	−0.10, −0.05	−0.24, −0.26	−0.13, −0.13	−0.33, −0.39	−0.26, −0.24	−0.32, −0.28	−0.12, −0.12	−0.18, −0.18	−0.20, −0.17
20. Latvia	−0.22, −0.22	−0.62, −0.62	−0.08, −0.11	−0.75, −0.72	−0.44, −0.44	−0.59, −0.59	−0.51, −0.49	−0.69, −0.67	−0.45, −0.43
21. Lithuania	−0.22, −0.22	−0.53, −0.54	−0.11, −0.12	−0.49, −0.47	−0.50, −0.47	−0.55, −0.52	−0.37, −0.37	−0.44, −0.44	−0.38, −0.36
22. Luxembourg	-	−0.66, −0.66	−0.59, −0.59	−0.69, −0.68	−0.50, −0.55	−0.33, −0.27	−0.53, −0.64	−0.79, −0.79	−0.44, −0.44
23. Mexico	−0.40, −0.41	−0.14, −0.16	−0.69, −0.70	−0.57, −0.56	−0.45, −0.41	−1.72, −1.63	-	−0.51, −0.49	−0.67, −0.65
24. Netherlands	−0.11, −0.12	−0.29, −0.29	−0.05, −0.08	−0.38, −0.41	−0.46, −0.48	−0.33, −0.33	−0.11, −0.12	−0.30, −0.30	−0.23, −0.24
25. New Zealand	−0.40, −0.40	−0.69, −0.71	−0.22, −0.23	−0.77, −0.80	−0.04, −0.09	−0.63, −0.66	−0.45, −0.51	−0.96, −0.93	−0.59, −0.58
26. Norway	−0.50, −0.45	−0.83, −0.84	−0.30, −0.31	−0.92, −0.92	−0.50, −0.58	−0.96, −0.94	−0.28, −0.36	−0.75, −0.76	−0.88, −0.78
27. Poland	−0.21, −0.16	−0.66, −0.67	−0.36, −0.36	−0.82, −0.78	−1.08, −0.97	−0.73, −0.73	−0.43, −0.41	−0.96, −0.96	−0.57, −0.52
28. Portugal	-	−0.68, −0.69	−0.23, −0.29	−0.30, −0.24	−0.70, −0.69	−0.19, −0.09	−0.57, −0.58	−0.51, −0.51	−0.51, −0.55
29. Slovakia	−0.22, −0.20	−0.20, −0.14	−0.36, −0.35	−0.66, −0.64	−0.55, −0.58	−0.60, −0.57	−0.68, −0.62	−0.52, −0.54	−0.46, −0.44
30. Slovenia	−0.19, −0.19	−0.48, −0.50	−0.18, −0.18	−0.46, −0.49	−0.40, −0.39	−0.41, −0.44	−0.35, −0.35	−0.28, −0.29	−0.26, −0.25
31. South Africa	−0.18, −0.17	−0.46, −0.50	−0.18, −0.20	−0.35, −0.38	−0.35, −0.34	−0.41, −0.43	−0.27, −0.31	−0.53, −0.54	−0.34, −0.30
32. Spain	−0.08, −0.08	−0.26, −0.28	−0.09, −0.11	−0.27, −0.29	−0.16, −0.17	−0.28, −0.30	−0.03, −0.05	−0.22, −0.23	−0.21, −0.22
33. Sweden	−0.24, −0.24	−0.75, −0.74	−0.11, −0.08	−0.90, −0.86	−0.11, −0.16	−0.66, −0.67	-	−0.34, −0.36	−0.48, −0.47
34. Switzerland	-	−1.58, −1.57	−1.32, −1.30	−1.61, −1.61	−1.46, −1.43	−1.44, −1.44	−1.81, −1.79	−1.76, −1.78	−1.08, −1.08
35. Turkey	−0.05, −0.05	−0.08, −0.03	−0.08, −0.08	−0.28, −0.29	−0.23, −0.24	−0.28, −0.30	−0.25, −0.24	−0.11, −0.11	−0.18, −0.16
36. UK	−0.35, −0.34	−0.88, −0.87	−0.31, −0.31	−0.97, −0.99	−0.83, −0.83	−0.97, −1.00	−0.42, −0.49	−0.82, −0.82	−0.70, −0.69
37. United States	−0.16, 0.00	−0.57, −0.57	−0.33, −0.32	−0.72, −0.78	−0.42, −0.42	−0.78, −0.76	−0.39, −0.44	−0.49, −0.49	−0.55, −0.56
Mean	−0.32, −0.22	−0.51, −0.51	−0.32, −0.26	−0.58, −0.58	−0.46, −0.46	−0.59, −0.59	−0.42, −0.44	−0.52, −0.52	−0.45, −0.45
Median	−0.20, −0.16	−0.46, −0.47	−0.18, −0.21	−0.52, −0.54	−0.43, −0.42	−0.54, −0.52	−0.36, −0.37	−0.44, −0.44	−0.45, −0.44
SD	0.40, 0.27	0.40, 0.39	0.49, 0.25	0.35, 0.33	0.33, 0.34	0.40, 0.39	0.42, 0.44	0.39, 0.39	0.29, 0.29

Note: The compensated own-price elasticity of good i in country c is π_{ii}^c/w_{ic}, where $\pi_{ii}^c = \phi^c\theta_i^c(1-\theta_i^c)$ is the own-Slutsky coefficient with ϕ^c the income flexibility and θ_i^c the marginal share of i in c; and $w_{ic}$is the budget share of i in c. The elasticities are evaluated with: (i) the estimates of ϕ^c from column 11 of Table 6; and (ii) θ_i^c specified as $w_{ic} + \beta_i^c$, with β_i^c-estimates from columns 2–16 of Table 6; and w_{ic} equal to (a) the mean of w_{ic}, for the elasticity value in the first entry of each cell; and (b) the last value of w_{ic} in the sample period, for the second entry. '-' refers to excluded entries corresponding to unusual income elasticity estimates.

Table 10. Comparison of the Income elasticities in 2 time periods, 6 Commodities, 22 countries[a].

	Country	Data period		Food		Clothing		Housing		Durables		Health		Recreation	
	(1)	SS(2003)	Current	SS(2003)	Current	SS(2003)	Current	SS(2003)	Current	SS(2003)	Current	SS(2003)	Current	SS(2003)	Current
1	Australia	1960–1996	1995–2015	0.43	0.34	1.27	0.83	0.34	0.77	2.57	0.80	1.11	1.24	2.04	1.33
2	Austria	1964–1996	1970–2014	0.65	0.28	1.54	0.62	0.49	0.88	1.30	0.97	0.40	0.82	0.99	1.53
3	Belgium	1960–1996	1995–2013	0.43	−0.27	1.20	−0.36	0.54	0.77	1.75	1.44	0.81	1.18	0.77	1.66
4	Canada	1960–1995	1970–2015	0.72	0.23	1.48	1.26	0.43	0.41	1.23	1.63	1.28	1.14	1.36	1.43
5	Denmark	1966-1995	1996-2014	0.45	0.33	1.61	0.67	0.36	0.18	1.50	1.43	0.27	0.56	0.97	1.53
6	Finland	1960–1996	1995–2015	0.61	0.40	1.47	1.36	0.05	0.37	1.63	1.48	0.51	1.05	1.08	1.45
7	France	1964–1996	1993–2015	0.42	0.40	0.83	0.58	0.64	1.17	1.85	1.07	0.15	1.19	1.55	1.39
8	Germany	1961–1994	1995–2014	0.67	0.40	1.45	0.67	0.25	0.39	1.50	1.44	0.63	0.89	0.27	1.75
9	Greece	1961–1995	1995–2014	0.52	0.73	2.23	1.68	0.56	0.35	1.20	1.91	1.89	1.87	1.75	1.43
10	Iceland	1977–1996	1975–2015	0.34	0.40	1.26	0.99	0.00	1.20	2.01	0.87	0.62	0.38	0.95	0.63
11	Ireland	1973–1996	1996–2014	0.44	0.73	2.04	0.87	0.42	1.81	1.72	0.74	0.83	0.55	1.74	0.81
12	Italy	1964–1996	1995–2015	0.70	0.73	1.77	1.31	0.49	0.17	1.60	1.46	1.01	1.00	0.85	1.85
13	Japan	1970–1996	1994–2013	0.79	0.48	1.80	0.67	0.30	0.40	1.28	1.32	0.81	−0.46	0.77	1.61
14	Luxembourg	1970–1991	1995–2015	0.43	2.11	1.11	0.82	0.92	0.86	1.63	0.87	2.03	0.60	−0.09	0.40
15	Netherlands	1952–1996	1995–2015	0.47	0.42	1.85	1.08	0.55	0.18	1.90	1.50	0.69	1.77	1.01	1.38
16	New Zealand	1983–1995	1987–2014	0.44	0.76	1.61	1.17	1.09	0.39		1.34	0.85	0.06		1.15
17	Norway	1964–1996	1995–2015	0.63	0.64	1.34	0.98	−0.03	0.36	1.50	1.09	1.68	0.56	1.19	1.21
18	Spain	1964–1996	1995–2014	0.84	0.41	0.78	1.30	0.69	0.43	0.92	1.32	1.71	0.77	1.16	1.44
19	Sweden	1963–1996	1995–2014	0.74	0.51	1.64	1.56	0.36	0.22	1.73	1.93	0.18	0.21	1.42	1.55
20	Switzerland	1961–1994	1998–2014	0.96	1.35	1.42	0.96	0.40	1.03	1.44	0.98	0.79	0.98	1.23	0.93
21	UK	1961–1996	1975–2014	0.61	0.45	1.00	1.16	0.30	0.42	1.66	1.28	0.80	1.04	0.87	1.39
22	United States	1961–1996	2000–2014	0.37	0.27	1.33	0.99	0.52	0.61	1.59	1.30	0.33	0.79	1.16	1.48
	Mean			0.58	0.55	1.46	0.96	0.44	0.61	1.60	1.28	0.88	0.83	1.10	1.33

[a] SS (2003) refers to Selvanathan and Selvanathan (2003). Only countries and commodities that are common to both studies are considered here.

Table 11. Comparison of the own-price elasticities in 2 time periods, 6 commodities, 22 countries[a].

	Country	Data period		Food		Clothing		Housing		Durables		Health		Recreation	
	(1)	SS(2003)	Current	SS(2003)	Current	SS(2003)	Current	SS(2003)	Current	SS(2003)	Current	SS(2003)	Current	SS(2003)	Current
1	Australia	1960–1996	1995–2015	−0.21	−0.13	−0.62	−0.33	−0.18	−0.32	−1.12	−0.32	−0.56	−0.48	−0.96	−0.47
2	Austria	1964–1996	1970–2014	−0.14	−0.12	−0.34	−0.28	−0.12	−0.42	−0.30	−0.42	−0.10	−0.37	−0.24	−0.59
3	Belgium	1960–1996	1995–2013	−0.03		−0.07		−0.03	−0.07	−0.09	−0.07	−0.05	−0.05	−0.05	−0.07
4	Canada	1960–1995	1970–2015	−0.37	−0.08	−0.80	−0.41	−0.24	−0.51	−0.66	−0.51	−0.74	−0.38	−0.75	−0.43
5	Denmark	1966–1995	1996–2014	−0.23	−0.16	−0.84	−0.35	−0.19	−0.70	−0.76	−0.70	−0.15	−0.29	−0.52	−0.67
6	Finland	1960–1996	1995–2015	−0.06	−0.18	−0.15	−0.61	−0.01	−0.66	−0.16	−0.66	−0.05	−0.49	−0.11	−0.59
7	France	1964–1996	1993–2015	−0.15	−0.16	−0.32	−0.25	−0.23	−0.44	−0.62	−0.44	−0.06	−0.51	−0.57	−0.54
8	Germany	1961–1994	1995–2014	−0.22	−0.11	−0.49	−0.19	−0.10	−0.39	−0.50	−0.39	−0.24	−0.26	−0.10	−0.44
9	Greece	1961–1995	1995–2014	−0.11	−0.11	−0.45	−0.19	−0.13	−0.39	−0.28	−0.39	−0.45	−0.26	−0.42	−0.44
10	Iceland	1977–1996	1975–2015	−0.10	−0.30	−0.36	−0.76	0.00	−0.66	−0.52	−0.66	−0.20	−0.30	−0.28	−0.47
11	Ireland	1973–1996	1996–2014	−0.09	−1.57	−0.41		−0.09		−0.35	−1.74	−0.19	−1.33	−0.35	−1.88
12	Italy	1964–1996	1995–2015	−0.05	−0.33	−0.13	−0.62	−0.04	−0.69	−0.13	−0.69	−0.09	−0.51	−0.07	−0.85
13	Japan	1970–1996	1994–2013	−0.26	−0.27	−0.62	−0.40	−0.11	−0.75	−0.47	−0.75	−0.30		−0.28	−0.81
14	Luxembourg	1970–1991	1995–2015	−0.19		−0.48	−0.53	−0.36	−0.49	−0.65	−0.49	−0.84	−0.50	0.04	−0.55
15	Netherlands	1952–1996	1995–2015	−0.05	−0.11	−0.16	−0.29	−0.06	−0.38	−0.17	−0.38	−0.07	−0.46	−0.10	−0.33
16	New Zealand	1983–1995	1987–2014	−0.56	−0.40	−2.04	−0.69	−1.02	−0.77		−0.77	−1.02	−0.04		−0.63
17	Norway	1964–1996	1995–2015	−0.30	−0.50	−0.70	−0.83	0.02	−0.92	−0.76	−0.92	−0.92	−0.50	−0.62	−0.96
18	Spain	1964–1996	1995–2014	−0.12	−0.08	−0.14	−0.26	−0.12	−0.27	−0.16	−0.27	−0.30	−0.16	−0.20	−0.28
19	Sweden	1963–1996	1995–2014	−0.16	−0.24	−0.39	−0.75	−0.09	−0.90	−0.41	−0.90	−0.05	−0.11	−0.33	−0.66
20	Switzerland	1961–1994	1998–2014	−0.34		−0.65	−1.58	−0.19	−1.61	−0.65	−1.61	−0.34	−1.46	−0.54	−1.44
21	UK	1961–1996	1975–2014	−0.06	−0.35	−0.11	−0.88	−0.03	−0.97	−0.17	−0.97	−0.09	−0.83	−0.09	−0.97
22	United States	1961–1996	2000–2014	−0.12	−0.16	−0.40	−0.57	−0.16	−0.72	−0.48	−0.72	−0.11	−0.42	−0.36	−0.78
	Mean			−0.18	−0.28	−0.49	−0.54	−0.16	−0.62	−0.45	−0.67	−0.31	−0.46	−0.33	−0.68

a SS (2003) refers to Selvanathan and Selvanathan (2003). Only countries and commodities that are common to both studies are considered here.

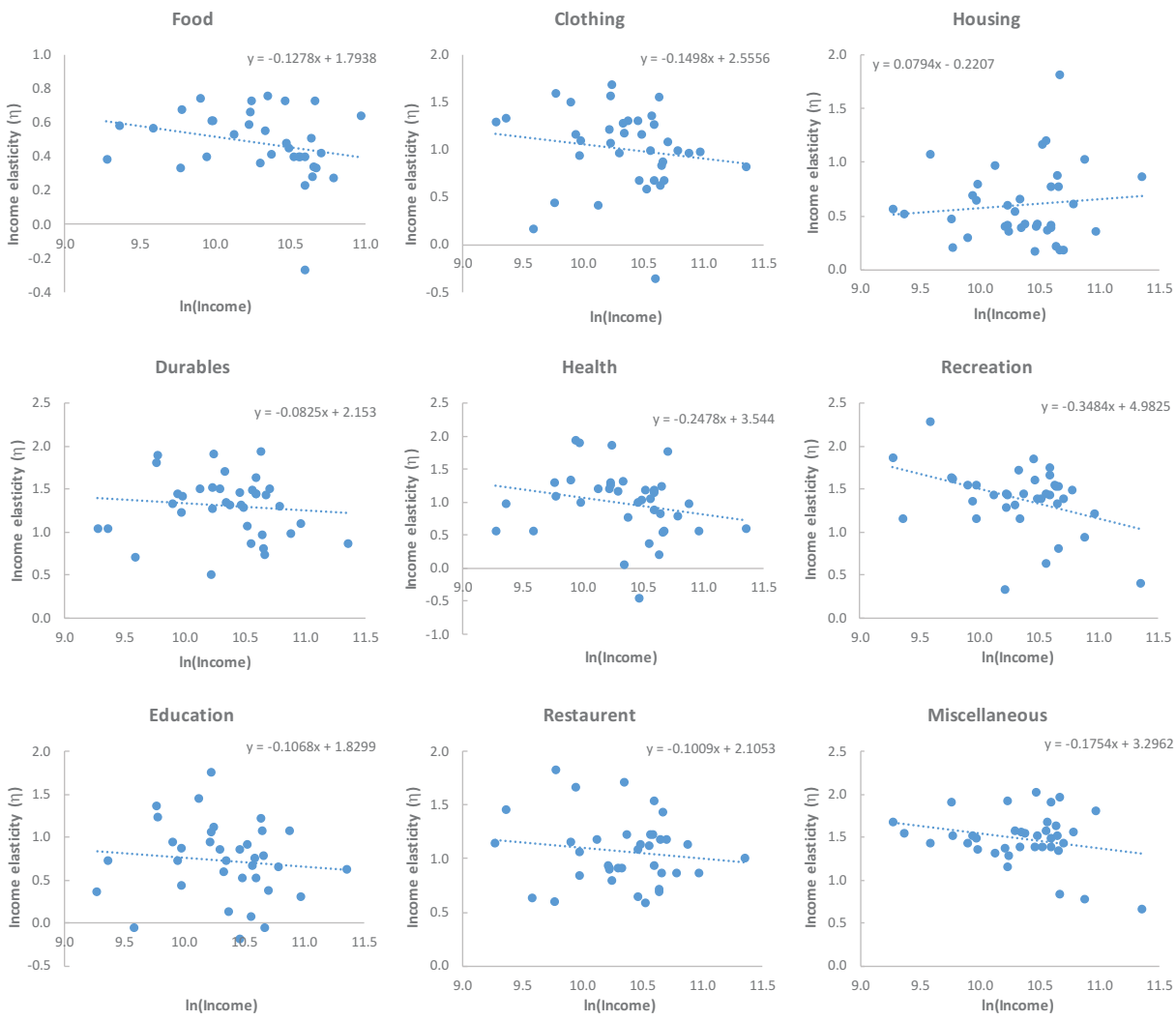

Figure 7. Income elasticity vs ln(Income), all countries combined, 9 commodity groups.

on the right-hand side of equation (6) and then subtracting $\bar{w}_{it}Dq_{it}$ from both sides gives:

$$\bar{w}_{it}(Dq_{it} - DQ_t) = \beta_i DQ_t + \phi(\bar{w}_{it} + \beta_i)\left[Dp_{it} - \Sigma_{j=1}^{n}(\bar{w}_{jt} + \beta_j)Dp_{jt}\right] + \varepsilon_{it}. \quad (7)$$

Thus, β_i is an income coefficient that satisfies $\Sigma_{i=1}^{n}\beta_i = 0$. As this is an allocation system with $\Sigma_{i=1}^{n}\bar{w}_{it}(Dq_{it} - DQ_t) = 0$, one of the n equations is redundant and can be dropped. We estimate the demand system (7) for i. = 1, 2, …, (n-1) under the assumptions that the disturbance term is independence over time with $\mathrm{E}(\varepsilon_{it}) = 0$ and $\mathrm{E}(\varepsilon_{it} \cdot \varepsilon_{jt}) = \sigma_{ij}$, a constant.

Since demand system (7) is non-linear in the parameters, we estimate (7) by maximum likelihood (ML). We use the Monte-Carlo simulation test proposed in S. Selvanathan (1987) to test for preference independence. Under this experiment, using 1000 simulations, if the rank of the test statistic is between 991 and 1000, the null hypothesis is rejected at the 1 percent level of significance and if the rank of the test statistic is between 951 and 1000, we reject preference independence at the 5% level of significance. Table 6 gives the results for testing preference independence for the 37 countries. As can be seen, in general, except for three countries (Canada, France and the USA) preference independence is acceptable. In what follows, we use the preference independent version of the demand system (7) for estimation and then to calculate the income and price elasticities based on equations (4) and (5).

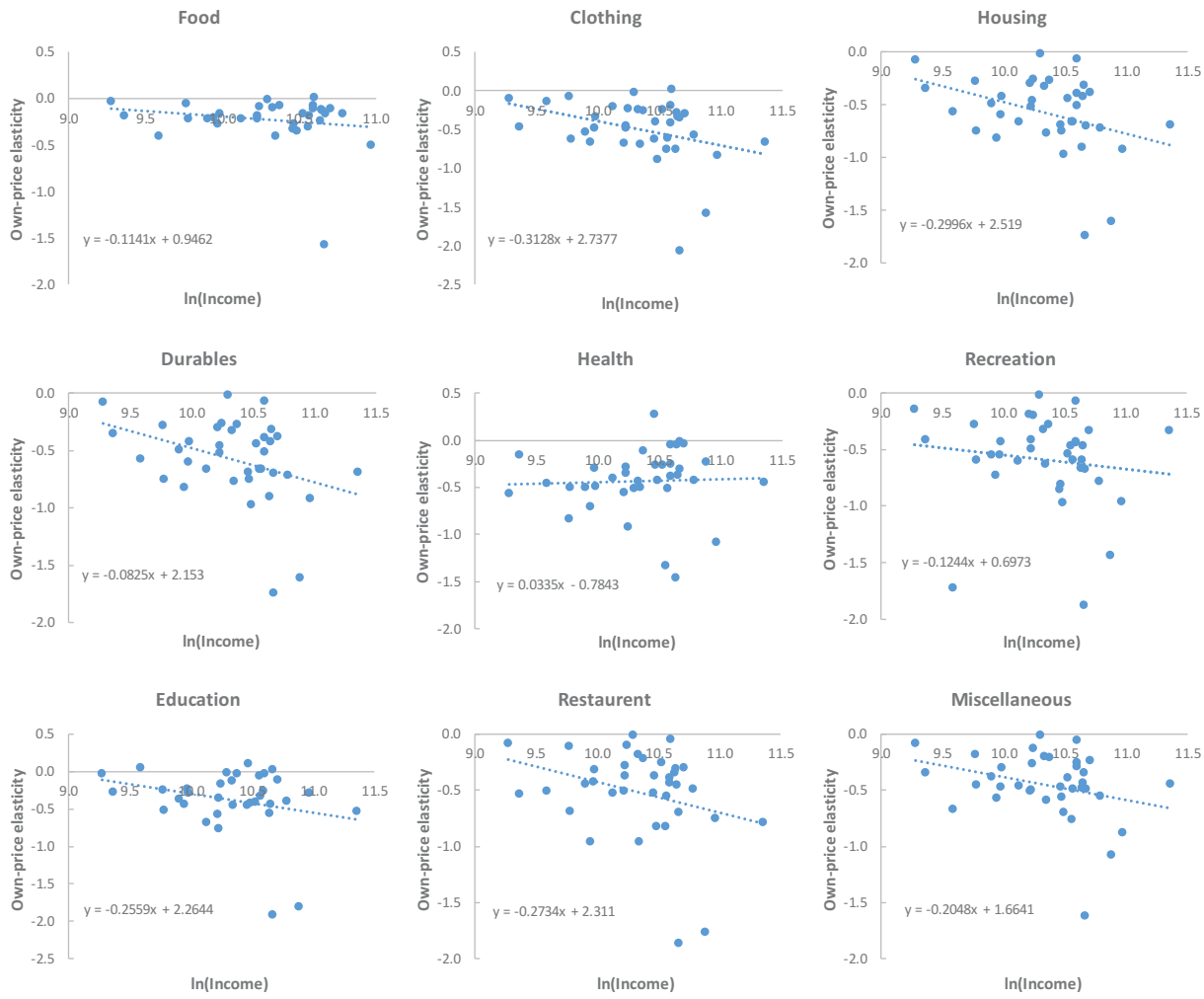

Figure 8. Own-price elasticity vs ln(Income) all countries combined, 9 Commodity Groups.

Table 7 presents the estimates of the demand system (7) under preference independence. As can be seen, on average, the beta coefficients for food, housing and education are negative indicating that, for OECD consumers, these three commodities are necessities and the other commodities are luxuries. The estimate of income flexibility given in the last column of Table 7 is negative as it should be with an average of −0.54 which is well in agreement with previous findings in single and cross-country consumptions studies (for example see, Theil and Clements 1987; Selvanathan 1993; Selvanathan and Selvanathan 2003).

V. The demand elasticities

Table 8 presents the income elasticities for the OECD countries implied by the Table 7 estimates. The last three rows of the table give the mean, median and standard deviation for each commodity. As can be seen, the median income elasticity for food and housing across OECD countries is less than one, indicating that they are necessities. The income elasticities for clothing, health and restaurants are close to one indicating that they are near luxuries. The income elasticity for durables and recreation are larger than one indicating that they are strong luxuries. In our model, there are only two constraints on the income elasticities. The first is the logical requirement that their budget-share weighted average is unity, which follows from the budget constraint. This is an aggregation constraint. The second constraint is that under preference independence, the income elasticities most all be positive. Our estimates automatically satisfy the aggregation constraint, but some violate

positivity. Looking at column 2 of Table 8, it can be seen that the food income elasticity for Belgium is negative; additionally, for three other countries, the elasticity is greater than one, which violates Engel's law. There are a few other instances for other commodities where the elasticities take unexpectedly high/low values. To a certain extent, some outlying values of this nature are to be expected in a large-scale application.

To highlight the outliers and the differing dispersion of the income elasticities, we present in Figure 6, box plots for each commodity which contain some of the unexpected elasticities, namely food, clothing, housing, health and education. The four 'odd' cases for food mentioned above are clearly evident as are the others for the remaining commodities. Note also the high dispersion for health and education; this could reflect different methods of financing these goods and their different treatment in the national accounts across countries. Whatever the reason for the unusual or outlying values, to be conservative, in what follows we shall exclude those cases in which the income elasticities either lie outside the fences in the box plot or do not satisfy required conditions.

Table 9 presents the own-price elasticities. As can be seen, for almost all countries, the own price elasticities are negative as they should be and less than one in absolute value[6], indicating that the demand for all goods are price inelastic.

In Tables 10 and 11, for comparison, we present the income and price elasticities from our current study (which uses mostly data from 1995 to 2015) and one of the earlier cross-country study (mostly data from 1960 to 1996) by Selvanathan and Selvanathan (2003). On average, across countries, there is not much difference in the income elasticities (except clothing) between the two studies. However, there are noticeable differences visible on the own-price elasticities between the two studies. This means that, in general, on average, consumption patterns in OECD countries have not changed much with changes in their income level but consumption is responsive more to price changes.

VI. Further analysis of the elasticities

Below, we further investigate the relationship between the income and price elasticities and consumer income. As seen in Table 2, the per capita GDP across the OECD countries vary widely, ranging from $85,621 for Luxembourg to $34,893 for Italy and to $10,680 for Colombia. For example, for food, it is expected that income elasticity for food, a necessity, will decline with increasing level of income. Below, in Figure 7, we plot the income elasticity for each of the nine commodities against log of per capita income. As can be seen, for all commodities (except housing), the points are scattered around a negatively sloped line, indicating that the income elasticity generally declines with increasing income.

We now plot for each commodity, in Figure 8, the own-price elasticities against log of per capita income. As can be seen, in all plots, the points are scattered around a negatively sloped straight line, indicating that when income increases, own-price elasticity also increases in absolute value. This means that for poor consumers, demand for any good is demand inelastic while for rich consumers, demand is price elastic.

VII. Conclusion

In this paper we have estimated for 37 OECD countries, the income and price elasticities of nine aggregate commodities, namely, (1) Food, (2) Clothing, (3) Housing, (4) Durables, (5) Health, (6) Recreation, (7) Education, (8) Restaurants and (9) Miscellaneous. For most countries, we have used annual data for the period 1995–2015. While demand homogeneity is acceptable for most countries, it is not the case with symmetry. We have identified a number of empirical regularities supported by the OECD consumption data. We investigated one of the special form of consumer's utility structure, preference independence and found that in almost 90 percent of countries, the data support this form. We have presented a tabulation of income and own-price elasticities for the 9 commodities and 37 countries, which

[6]Ireland and Switzerland values seem to be exception to this.

may be a convenient resource for other researchers. The income elasticity estimates reveal that across OECD countries, food and housing can be classified as necessities (income elasticity < 1); clothing, health and restaurants are near luxuries (income elasticity is close to one); and durables, recreation and education are luxuries (income elasticity > 1). All nine own-price elasticities are negative and less than one in absolute value indicating that the demands are price inelastic. A comparison of the income and own-price elasticities from the current study with a earlier study reveals that, on average, across countries there is not many noticeable differences between the income elasticities, but more differences in the own-price elasticities. We have also investigated the relationship between income elasticities and income and found that in general, the elasticity decreases with increasing income of the countries. We also investigated the relationship between the own-price elasticities and income and found that, in general, there is a negative linear relationship between own-price elasticities and income. The income and price elasticities presented in this paper are useful inputs for economy-wide modelling, which in turn would be crucial for a host of microeconomic policy recommendations including public-utility pricing, optimal taxation, the measurement of distortion and the treatment of externalities at the individual as well as cross-country levels.

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Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

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